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ECA56 –Advanced Wireless Communication Systems / 5G / 6G Communication Systems

Ex.No.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mm Wave by Plotting Propagation Loss.

AIM:

To analyze the propagation loss characteristics of different frequency bands (Sub-1 GHz, Mid-band, and mm Wave) in wireless communication systems and compare their performance over varying distances using MATLAB.

Procedure :

1. Define frequency bands: Sub-1 GHz (0.9 GHz), Mid-band (3.5 GHz), and mmWave (28 GHz).
2. Set communication distance range (e.g., 0.1 km to 5 km).
3. Use Free Space Path Loss (FSPL) formula to compute propagation loss:

$PL(dB)=20\log10(d)+20\log10(f)+32.44$

4. Calculate propagation loss for each frequency band over varying distances.
5. Plot distance vs propagation loss curves for comparison.
6. Analyze how higher frequencies experience higher attenuation.
7. Compare coverage efficiency of each spectrum band.
8. Display results using MATLAB plots.

Objective 1 :

To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mm Wave frequency ranges using standard path loss models.

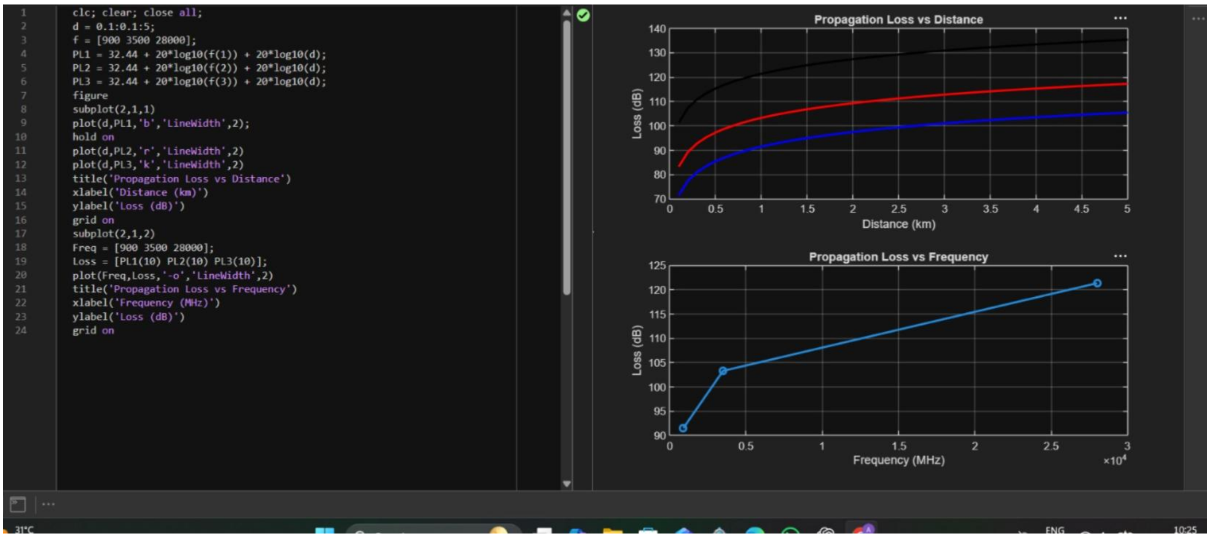
Program :

```
clc; clear; close all;

d = 0.1:0.1:5; % Distance (km)
```

```
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
% -------- Graph 2 (NO legend used to avoid error) --------
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```

Output :



Objective 2 :

To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.

Program :

```
clc; clear; close all;

d = 2; % fixed distance (km)

f = [900 3500 28000]; % MHz

PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% -------- Graph 1: Frequency vs Loss --------

subplot(2,1,1)

plot(f,PL,'-o','LineWidth',2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

% -------- Graph 2: Band Comparison --------

subplot(2,1,2)

bar(PL)
```

title('Comparison of Propagation Loss Across Bands')

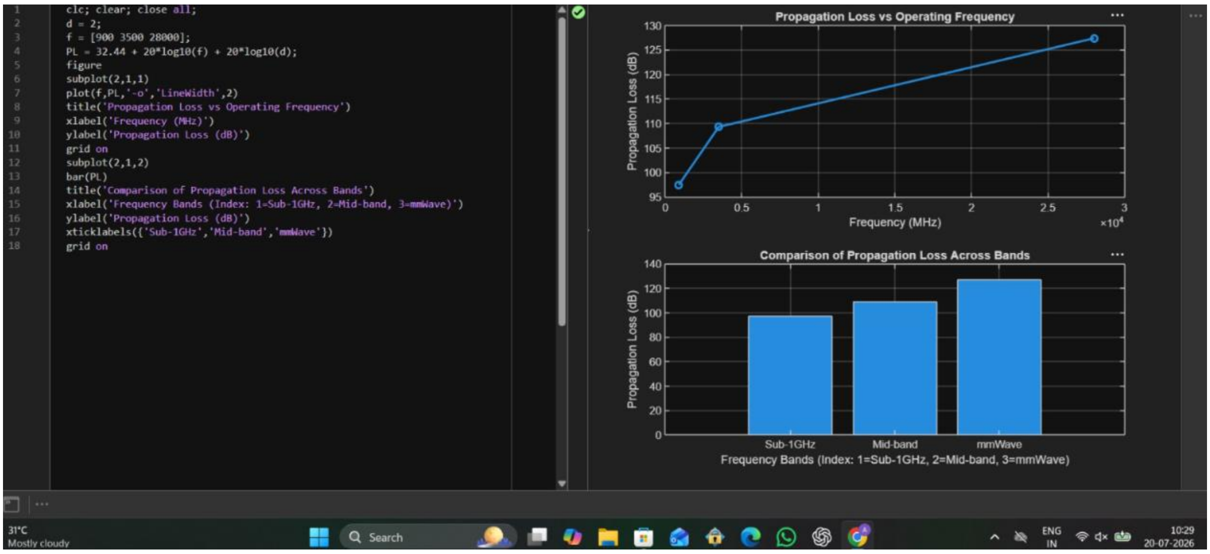
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

xticklabels({'Sub-1GHz','Mid-band','mmWave'})

grid on

Output :



Objective 3 :

To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

Program :

clc; clear; close all;

% Frequency bands (MHz)

f = [900 3500 28000];

% Distance (km)

d = 2;

% Propagation Loss (FSPL)

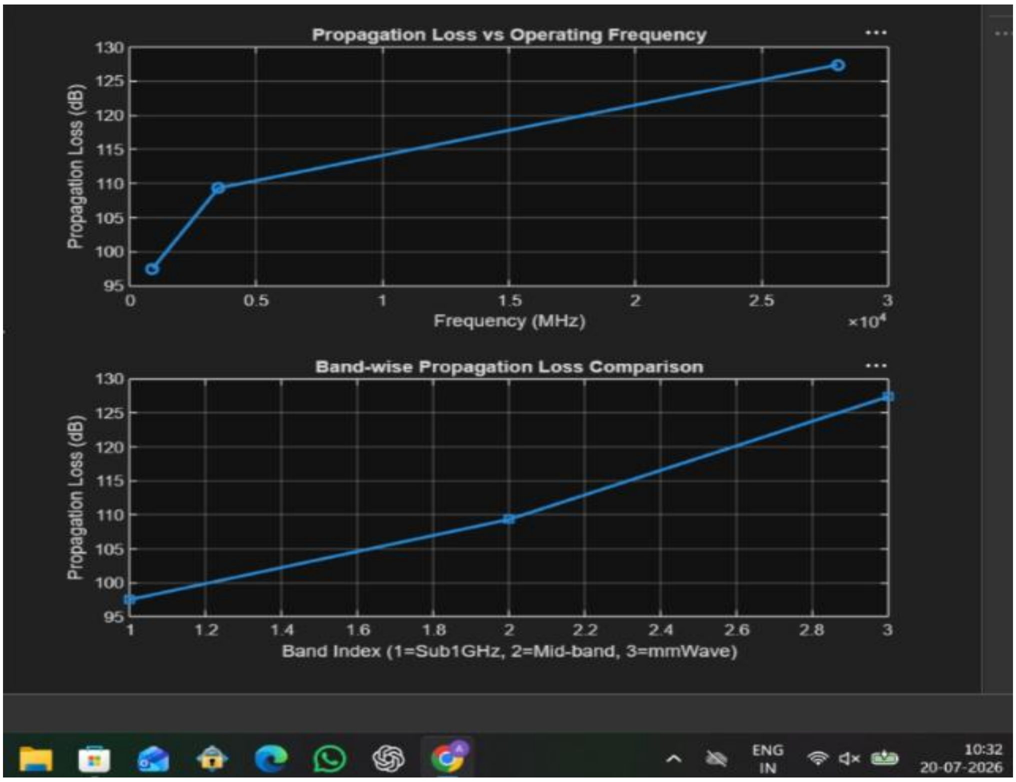
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% -------- Graph 1: Simple Line Plot (SAFE) --------

```
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
```

Output :



Result :

1. The propagation loss was successfully computed for Sub-1 GHz, Mid-band, and

mm Wave frequency bands using the Free Space Path Loss (FSPL) model.

2. It was observed that propagation loss increases with increase in operating frequency, with mm Wave showing the highest attenuation and Sub-1 GHz showing the lowest loss.

3. The comparative analysis of spectrum bands clearly demonstrates that lower frequency bands are more suitable for long-range communication, while higher frequency bands require advanced techniques like beamforming for efficient performance.